higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION: BOILERMAKING
NQF LEVEL 4

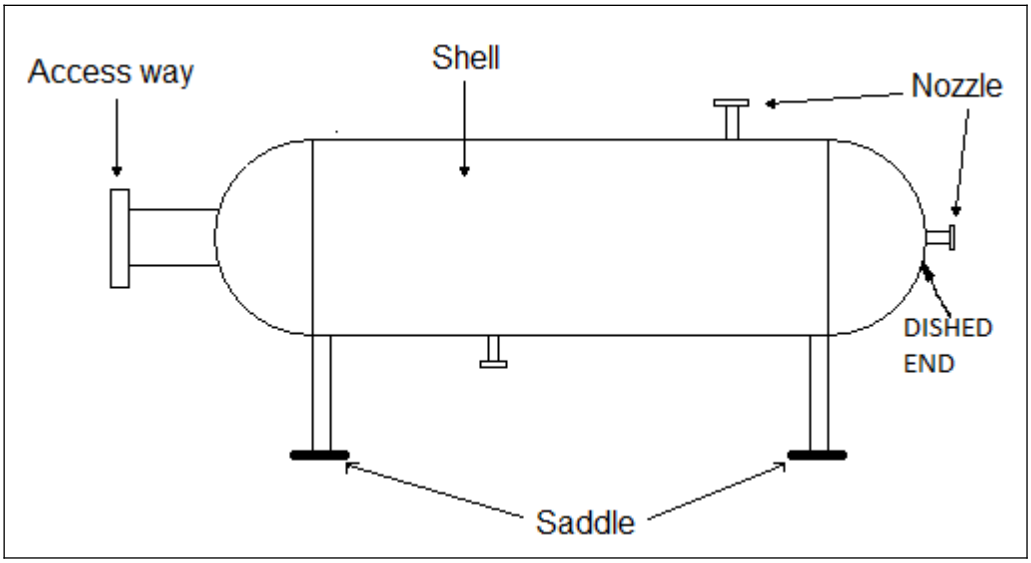
12 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1



(5 × 1) (5)

1.2

1.2.1	Weld all round
1.2.2	---------
1.2.3	⌒
1.2.4	M
1.2.5	Flame

(5 × 1) (5)

1.3

1.3.1	True
1.3.2	False
1.3.3	False
1.3.4	False
1.3.5	True
1.3.6	False
1.3.7	False
1.3.8	True
1.3.9	True
1.3.10	True

(10 × 1) (10)
[20]

QUESTION 2

2.1
2.1.1 D
2.1.2 E
2.1.3 A
2.1.4 B
2.1.5 C
(5 × 1) (5)

2.2 Compressive force (1)

2.3
2.3.1 H section
2.3.2 I section
2.3.3 Cap plate
2.3.4 Base plate
(4 × 1) (4)

2.4
- To stabilise the structure against sideways swaying
- To prevent the structure from buckling
- To prevent the structure from distorting (3)

2.5
- General safety when working on site
- Safety when handling machines, tools and equipment
- Safety clothing to protect you
- Safety in how a site is managed, how people communicate on site and how the site is kept safe and tidy (4)

2.6
- Bow shackle
- 'D' shackle (2 × 1) (2)

2.7 Gloves (1)
[20]

QUESTION 3

3.1
- Structural drawings
- Piping equipment arrangement drawings
- Section views
- Piping and instrumentation drawings (P&DI)
- Isometric drawings (ISO)
- Spool drawings (Any 5 × 1) (5)

3.2
- A cross valve
- A globe valve
- An angle valve
- A gate valve
- A check valve
- A 90° elbow
- A 45° elbow
- A coupling
- A reduce coupling
- A straight tee
- A union
- A cap
- A cross
- A plug (Any 5 × 1) (5)

3.3 $\sin 30° = \frac{\text{Opposite}}{\text{Hypotenuse}}$ ✓

$\text{Hypotenuse (travel)} = \frac{\text{Offset}}{\sin 30°}$ ✓

$= \frac{300 \text{ mm}}{0{,}5}$ ✓

$= 150 \text{ mm}$ ✓

The travel or length of pipe required is 150 mm✓ (5)

3.4
- The work you must do
- Where the work is
- When the work must be done
- How much time you have to complete the job
- How many people are needed to do the job
- The equipment that you have to work on
- The condition of the equipment
- Repairs that have been done
- If there was any time lost (Any 5 × 1) (5)

[20]

QUESTION 4

4.1 4.1.1
- Engineering and machining industry
- Fabrication industry
- Woodworking industry (3 × 1) (3)

4.1.2 Engineering and machining industry
- Milling machines
- Lathes
- Grinders (Any 1 × 1)

Fabrication industry
- Laser cutters
- Plasma cutters
- Shearing cutters
- Flame cutters
- Welding
- Forming machines
- Water jet machines
- Punching machines (Any 1 × 1)

Woodworking industry
- Lathes
- Routers
- Drilling machines
- Multi-head borers (Any 1 × 1) (3)

4.2
- It is labour intensive.
- Small holes must be drilled, and drill bit changed for bigger holes.
- Holes must be tapped.
- Holes must be reamed.
- Holes must be countersunk.
- An experienced machinist is needed.
- Each work piece must be marked accurately individually.
- This is a time-consuming process.
- Costly and could lead to inaccuracies.
- Quantities too small to warrant making a jig. (Any 7 × 1) (7)

4.3
- To use a CNC machine a simple program is written.
- An operator sets up the machine and it is ready for production.
- After the first part is machined, it is checked for accuracy and any adjustments can be made.
- Some machines even load the workpiece into position and clear away the scrap.
- Machines have turrets that hold the tools and the machine automatically accesses it.
- The program tells the machine which tool must be used. (Any 5 × 1) (5)

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4.4 4.4.1 An absolute movement is a movement to a position that was measured from a datum. You measure all the points from one fixed point.

4.4.2 An incremental movement is one made from the previous position that was reached. You measure this from the last position of the tool.

(2 × 2) (4)

4.5 4.5.1 The T-code tells the machine what tool station is to be placed in the spindle.

4.5.2 The S-code tells the machine at what speed the spindle must rotate in revolutions per minute.

4.5.3 The F-code is the feed rate and is either in millimetres per minute or inches per minute.

(3 × 2) (6)

4.6
- To see if there are any errors
- Alarm or error message comes on
- The error can be located and repaired (3 × 1) (3)

4.7
- Make sure that the cutter bodies are rigid and secure
- Make sure that the machine power is adequate for the programmed feed.
- Make sure the spindle head has zero play in the bearing.
- Make sure that the work piece fixtures are suitable for the work piece and that they are the quick-clamp / quick-release type.
- Make sure that the tool holders for both carbide and aluminium tips are rigid. (5)

4.8
- The type of metal being cut
- The speed and feed that is used
- The type of cutting tool that is used (3)

4.9 To check if the workpiece is within tolerance and complies with the drawing. (1)

[40]

TOTAL: 100

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